

Sushmita Paul

+880-188-170-1129 Mail LinkedIn GitHub Portfolio

EDUCATION

Bangladesh University of Engineering and Technology (BUET) <i>Bachelor of Science in Computer Science and Engineering</i> • CGPA: 3.80/4.00 • Relevant Coursework: Artificial Intelligence, Machine Learning, Software Engineering, Data Structures & Algorithms, Operating Systems, Computer Security, Computer Networks	Feb 2020 – Mar 2025 Dhaka, Bangladesh
Chittagong College <i>Higher Secondary Certificate</i> • GPA: 5.00/5.00	2017 – 2019 Chittagong, Bangladesh

HONORS & AWARDS

Dean's List Award	BUET
B.Sc. in Computer Science and Engineering (Honors)	BUET
Top Performer (Higher Secondary Certificate)	Chittagong Education Board

RESEARCH INTERESTS

Reliable, deployment-aware computer vision for video and aerial systems under resource and uncertainty constraints

PUBLICATIONS

1. **Paul, S., et al.** "Detection of Dengue Breeding Sites in Unplanned Urban Landscapes with Deep Learning from Remote Sensing," *BMC Public Health* (Springer Nature), 2025. [Under Review] (Co-first author)
2. **Paul, S., et al.** "Identification of Dengue Breeding Sites from Point-Cloud Images," *BEAR Summit 2025*, Bangladesh. [Digital Poster]

RESEARCH EXPERIENCE

Machine Learning for Systems & Video Streaming <i>Research Collaboration with Prof. Dr. Ch. Md. Rakin Haider</i> • Conducting literature-driven research on ML for systems optimization, covering OS scheduling, networked storage, fairness mechanisms, and streaming applications. • Identified gaps between algorithmic performance and real-world deployment constraints in video-centric systems. • Investigating reinforcement learning-based control policies that jointly optimize video bitrate and playback speed to maximize Quality-of-Experience (QoE) under dynamic network conditions while balancing energy efficiency and system constraints.	May 2025 – Present BUET
Computer Vision for Public Health <i>Undergraduate Thesis Research with Prof. Dr. Anindya Iqbal</i> • Developed an automated computer vision system to classify residential zones and detect dengue breeding sites from UAV-captured aerial imagery. • Systematically evaluated detection and segmentation models (YOLOv8, YOLOv11, SAM, SegGPT) on a custom-annotated dataset, selecting YOLO variants for optimal performance-efficiency trade-offs. • Engineered a geospatial analysis pipeline with sliding-window processing and Non-Maximum Merging to reduce false positives by 23% while improving computational efficiency. • Designed a distance-based search strategy accounting for spatial clustering patterns in dengue transmission. • Achieved 83.6% balanced accuracy and demonstrated 35% reduction in inspection costs, validating practical viability for public health deployment.	Nov 2023 – Jan 2025 BUET

PROFESSIONAL EXPERIENCE

Therap BD Ltd

Associate Machine Learning Engineer

Apr 2025 – Present

Dhaka, Bangladesh

- Optimized and deployed deep learning models on NVIDIA Jetson edge devices using TensorRT with FP16/INT8 quantization for real-time inference.
- Built end-to-end ML pipelines covering training, evaluation, and production deployment.
- Evaluated vision-language models for chart and document analysis, reproducing recent methods and analyzing accuracy-speed trade-offs for edge deployment.

TECHNICAL SKILLS

Programming & Systems

- **Languages:** Python, C/C++, Bash, Assembly
- **Tools:** Git, Docker, Linux
- **Compute:** SkyPilot, Azure, GPU Clusters, NVIDIA Jetson

Machine Learning & Computer Vision

- **Frameworks:** PyTorch, TensorFlow, scikit-learn
- **Model Efficiency:** TensorRT, quantization (FP16/INT8), pruning
- **Large Models:** LLM fine-tuning, benchmarking, RAG systems
- **Computer Vision:** Object detection (YOLO, Faster R-CNN), segmentation (SAM, SegFormer), OCR, video analytics, action recognition

Methods

- Transfer learning, ensemble methods, statistical modeling, experiment design

SELECTED PROJECTS

Medical Image Analysis: Lung Nodule Detection & Segmentation

Jan 2025

- Fine-tuned SAM2 and SwinUNETR for lung nodule segmentation and Faster R-CNN (ResNet50-FPN) for detection on the LIDC-IDRI dataset.
- Implemented point-prompt and sliding-window pipelines with hybrid loss functions, mixed-precision training, learning rate scheduling, and gradient accumulation.
- Achieved Dice score of 0.93, Precision of 0.96, and Recall of 0.89 while handling multi-annotator masks and clinical dataset challenges.

EduByte: AI-Enhanced E-Learning Platform

2024

- Built full-stack platform with personalized recommendations (collaborative filtering, content-based methods), Gemini-based conversational chatbot, and AI-powered examination system with automated question generation.
- Tech stack: React, Node.js, Express, PostgreSQL, RESTful API architecture.

TEACHING & OUTREACH

Private Mentor, Tutor for Mathematics and Physics

2020 – 2025

Dhaka, Bangladesh

- Tutored Higher Secondary Certificate students in STEM subjects with customized lesson plans tailored to individual learning styles. Mentored students in developing critical thinking and problem-solving approaches for competitive examinations.
- Delivered technical seminar at Islamic University of Technology (IUT) and developed promotional materials.